

Homework 3

Solve the following problems justifying all your answers. You should not use a graph calculator to justify any answer.

1. (a) Find $M > 0$ and $N > 0$, such that, if $|x - 1| < 1$; then,

$$|x - 5| < M \quad \text{and} \quad |x + 2| < N$$

- (b) Prove, using the (ϵ, δ) -definition of limit, that

$$\lim_{x \rightarrow 1} (x^3 - 4x^2 - 7x + 1) = -9$$

2. (a) Prove that if $|x - 2| < 1$; then $x + 2 > 3$. Conclude that, in this case, as $x + 2 > 0$; then, $|x + 2| > 3$.
 (b) Find $M > 0$, such that, if $|x - 2| < 1$; then, $\frac{1}{|x+2|} < M$.
 (c) Prove, using the (ϵ, δ) -definition of limit:

$$\lim_{x \rightarrow 2} \frac{4}{x + 2} = 1$$

3. Consider

$$f(x) = \begin{cases} \frac{x^3 - x^2 - x + 1}{x - 1} & \text{if } x < 1, \\ x^2 + ax + b & \text{if } 1 \leq x < \pi \\ 2 \cos(x - \frac{5\pi}{2}) & \text{if } x \geq \pi. \end{cases} \quad (1)$$

- (a) Find all the parameters $a, b \in \mathbb{R}$ for which the function f is continuous in $\mathbb{R} \setminus \{1, \pi\}$.
 (b) Find all the parameters $a, b \in \mathbb{R}$ for which the function f is continuous at $x = 1$ and at $x = \pi$.
 (c) Find all the parameters $a, b \in \mathbb{R}$ for which the function is continuous (everywhere).
 (d) Are there parameters $a, b \in \mathbb{R}$ for which the function f has an asymptote?

4. Consider $a \in [-\pi, 0]$ and the function

$$f(x) = \begin{cases} \sin x + a & \text{if } x \leq a, \\ \frac{x(x^2 - bx - ax + ab)}{x^2 - x - ax + a} & \text{if } a \leq x < 1 \\ e^{x-b} & \text{if } x \geq 1. \end{cases} \quad (2)$$

- (a) Find $a, b \in \mathbb{R}$ such that f is continuous.
 (b) For which parameters $a, b \in \mathbb{R}$ does the function f have horizontal or vertical asymptotes?
 (c) For each parameters $a, b \in \mathbb{R}$, find all the asymptotes of the function f .

Remark: When we don't specify the domain of where you need to prove the continuity of the function, we mean the maximal possible domain.